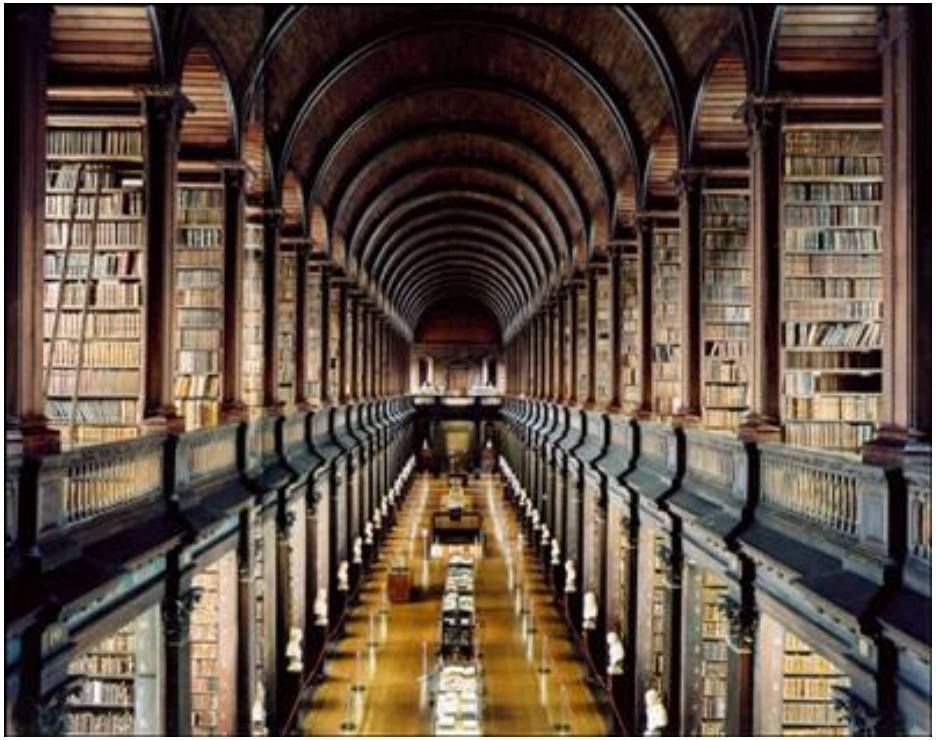
Text Analysis for Web Search



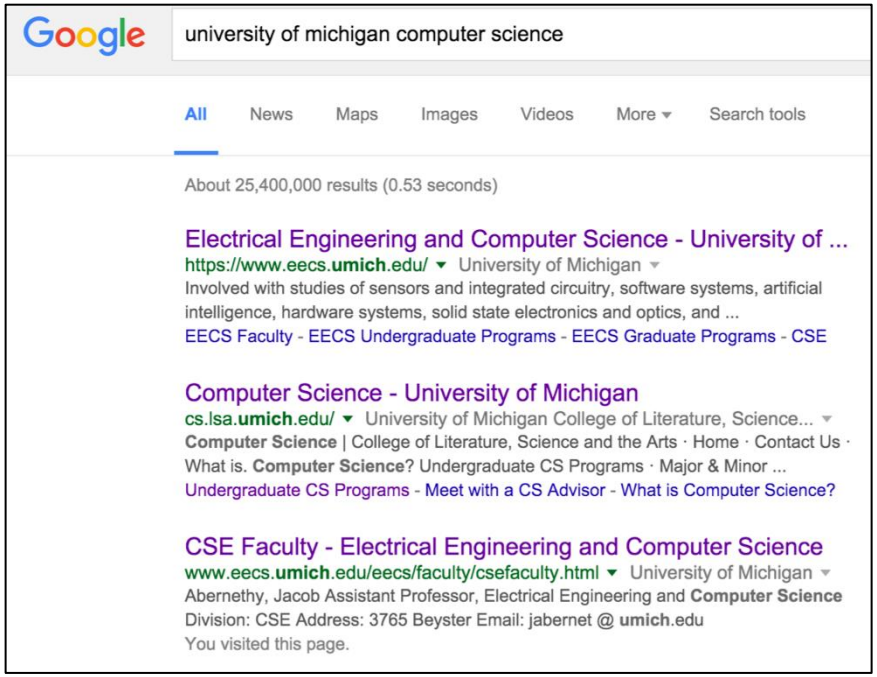
Slides by Raed Almomani, Andrew DeOrio, Raghavan et al.

Agenda

- Web search introduction
- Boolean retrieval
- Vector space model
- tf-idf

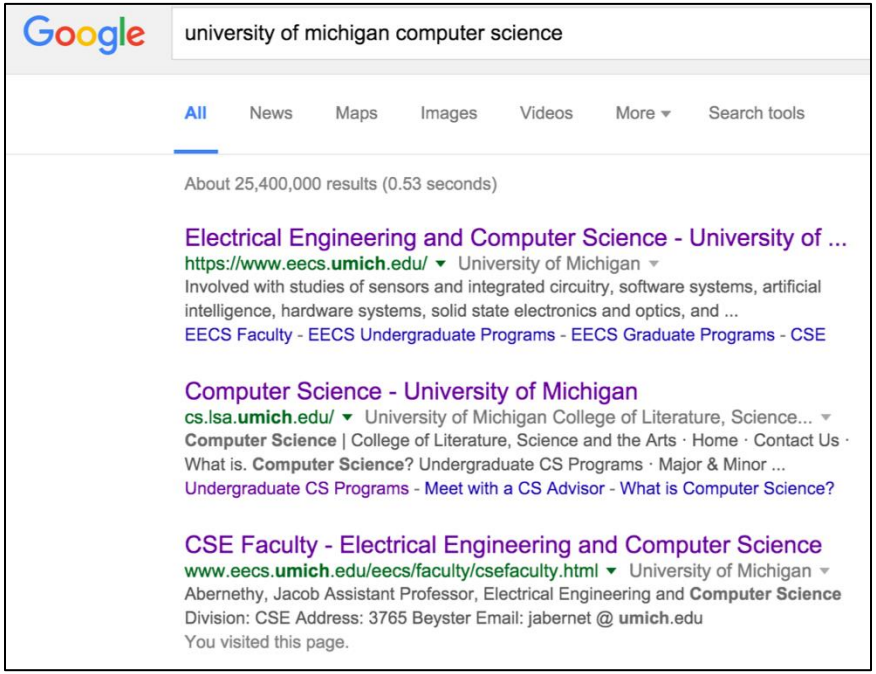
Web search

- ***Web search:*** find web pages relevant to user query
  - Example of *Information Retrieval (IR)*



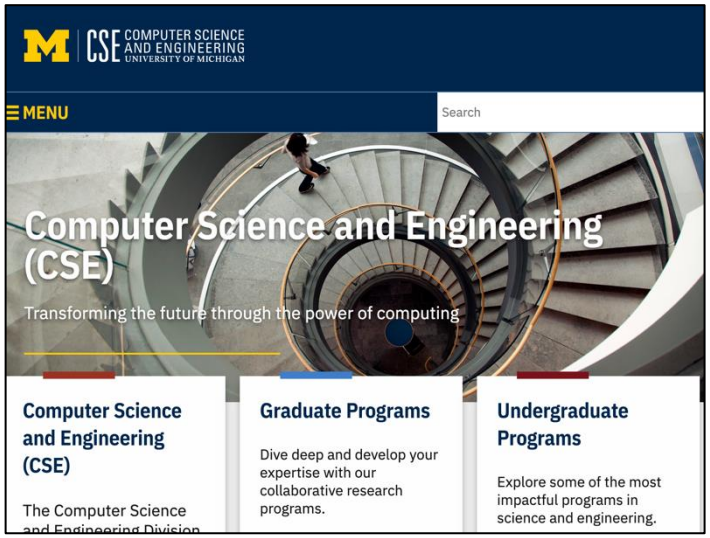
Anatomy of a web search

- *Query:* user input
- *Hits:* returned web pages
- *Rank:* order of results by relevance



Anatomy of a web search

- ***Document**:* one web page
- ***Collection**:* all documents
- ***Term**:* one word in a document
  - Or query



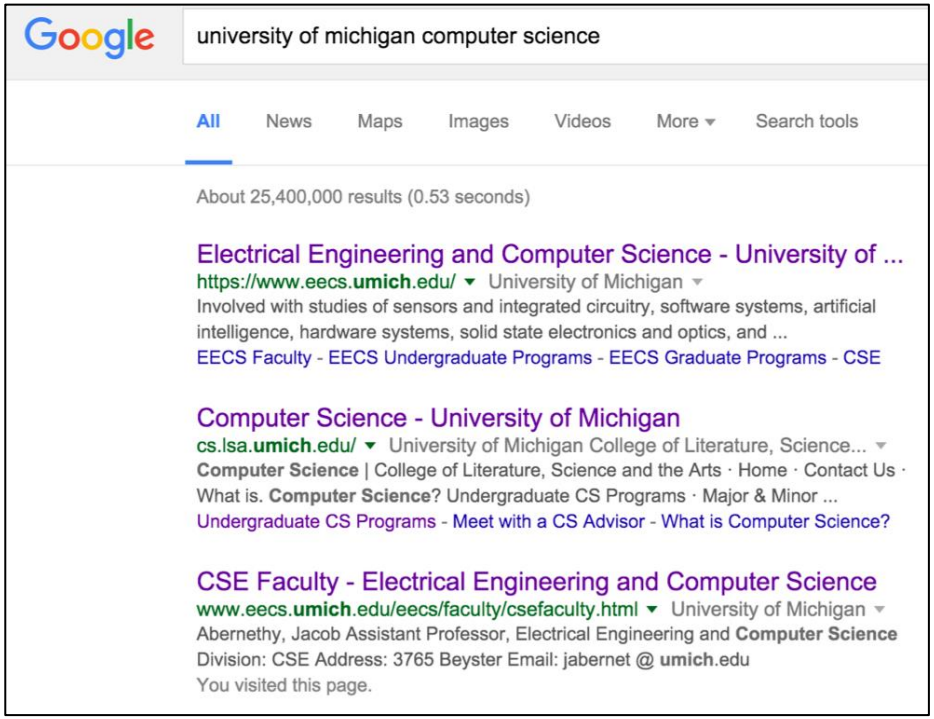
How search works

- Ahead of time (offline)
  - Download every web page
  - Build an index
- Query time
  - Read index
  - Rank results

- Problem: How do we rank results?

Ranking results

- Ranking is important
  - About 33% of users click the first result



- *Text analysis:* Rank using words in document
  - Today
- *Link analysis:* Rank using links between documents
  - Next time

Agenda

- What is search?
- **Boolean retrieval**
- Vector space model
- tf-idf

Boolean retrieval

- For each document, a query returns one of two outcomes
  - **TRUE** (document matches)
  - **FALSE** (document does not match)

Boolean retrieval

Kangaroo live in Australia and jump.

Cows live all over the world. Unlike kangaroo, they cannot jump.

Aluminum foil is shiny.

Index for Boolean retrieval

- **Inverted index**: map words to documents for efficient search.

Doc0

Kangaroo can jump.

Doc 1

Cows can not jump.

Term	Doc ID
kangaroo	0
can	0, 1
jump	0, 1
cow	1
not	1

Boolean search using inverted index

Term	Doc ID
kangaroo	0
can	0, 1
jump	0, 1
cow	1
not	1

Ranking search results

- Boolean queries only *include* or *exclude* documens from results
- When many documents match, we need to sort (rank) them

Agenda

- What is search?
- Boolean retrieval
- **Vector space model**
- tf-idf

Documents as vectors

- A document is a vector
- Each dimension represents a word
- # of dimensions: # of unique words in *all* documents

Boolean index to vectors

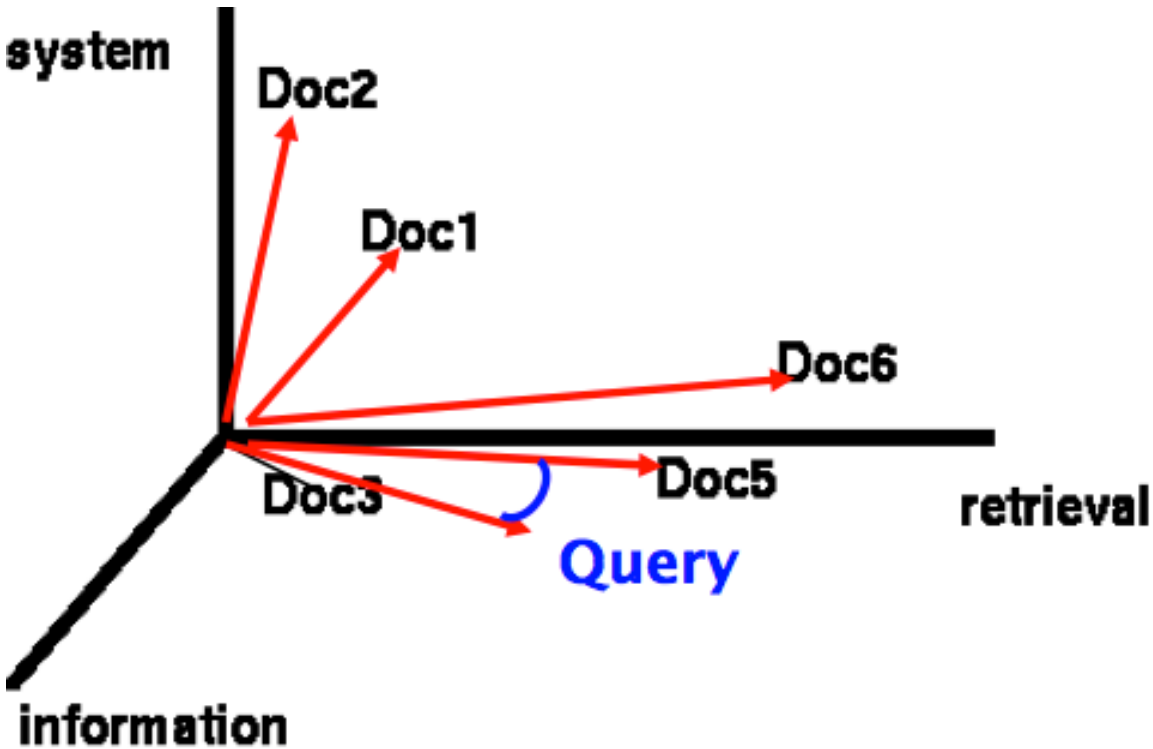
Term	Documents
kangaroo	0
can	0, 1
jump	0, 1
cow	1
not	1

Documents in n-D space

- Documents that are "close together" in space are close in meaning.

Documents in 3D space

- Term weights indicate length of document vector along a dimension



Vector space query model

1. Treat a query as a short document
2. Sort documents by increasing distance (decreasing similarity) to the query document
3. Easy to compute, both query and doc are vectors

Vector magnitude

- With Boolean search, the values in a vector (one for each term) are 0 or 1
- Next, we'll consider values [0, 1]

Q: Vector space dimensionality

- How many dimensions?

- Are vectors sparse or dense? Why?

A: Vector space dimensionality

- How many dimensions?
  - Dimension for every possible term (word)

- Are vectors sparse or dense? Why?
  - Sparse. Most documents do not use most words.

Agenda

- What is search?
- Boolean retrieval
- Vector space model
- **tf-idf**

Term frequency (tf)

- Which doc is a better match for the query "kangaroo”?
- One with a single mention of "kangaroo", or that mentions it 10 times?

Term frequency (tf)

- Which doc is a better match for the query "kangaroo", one with a single mention of "kangaroo", or a doc that mentions it 10 times?
- The doc that mentions it 10 times
- *Term frequency (TF):* the number of times the word appears in the document
- Higher is better

Term frequency (tf)

Kangaroo can jump. Kangaroo live in Australia.

Australia is in the southern hemisphere.

One of the longest flights is from London to Sydney, Australia.

Document frequency (DF)

- Document Frequency (DF) is the count of documents containing a specific term.

- In a collection of 100 documents, if 5 documents contain the term ‘AI’, then DF(AI) = 5

Document frequency (DF)

- Common terms with high DF (e.g., "data") appear in many documents and provide less unique value.
- Rare terms with low DF (e.g., "neural network") are more informative and distinctive.
- Lower is better

Inverse document frequency (idf)

- Inverse Document Frequency (IDF) provides high values for rare words, low values for common words
- *IDF(t) = N / DF(t)*
  - N = total # docs in collection *C*
  - *DT(t)* = # docs in *C* that contain *t*

<u>Example</u>

- If DF = 5 in a collection of 100 documents, IDF = 100/5 = 20.

Document frequency (df)

Kangaroo can jump. Kangaroo live in Australia.

Australia is in the southern hemisphere.

One of the longest flights is from London to Sydney, Australia.

Log inverse document frequency

- Query “AI book"
- What if “AI" appears 10 out of 10 million documents and "book" appears in 1 million out of 10 million?
- Is the term "book" 100,000 times less informative than “AI"?

Log inverse document frequency

- Query “AI book"
- What if “AI" appears 10 out of 10 million documents and "book" appears in 1 million out of 10 million?
- Is the term "book" 100,000 times less informative than “AI"?
  - Probably not.

- Solution: **log(N / n_k)**
- Sublinear function decreases the weight of “AI" compared to "book"
- Still monotonically increasing, so order is preserved

TF x IDF

- *Term frequency* high for common word in one document
- *Inverse Document Frequency* high for rare word in collection
- *Term-Frequency* x *Inverse-Document-Frequency* high for common word in one document that is rare in the collection
- $\boldsymbol{W_{ik} = tf_{ik} * \log(N / df(t))}$
  - T_k = term k in document D_i
  - tf_{ik} = freq of term T_k in doc D_i
  - N = total # docs in collection C
  - $DT(t)$ = # docs in C that contain t

TF-IDF normalization

- Imagine two documents about kangaroos
- Document 1 mentions "kangaroo" 100 times. Total length 1000 words.
- Document 2 mentions "kangaroo" 200 times. Total length 10,000 words.

- Which document should have the greater weight?
  - Avoid giving longer doc higher weight just because they are long
  - Need to normalize

TF-IDF normalization

- Normalize term weights
  - Normalize to vector length

$$w_{ik} = \frac{tf_{ik} \log(N/n_k)}{\sqrt{\sum_{k=1}^{t} (tf_{ik})^2 [\log(N/n_k)]^2}}$$

- Some references use non-normalized tf-idf
  - $w_{ik} = tf_{ik} \log(N/n_k)$

Vector space similarity

- Boolean vectors contain only 0 or 1 depending on whether they contain a word

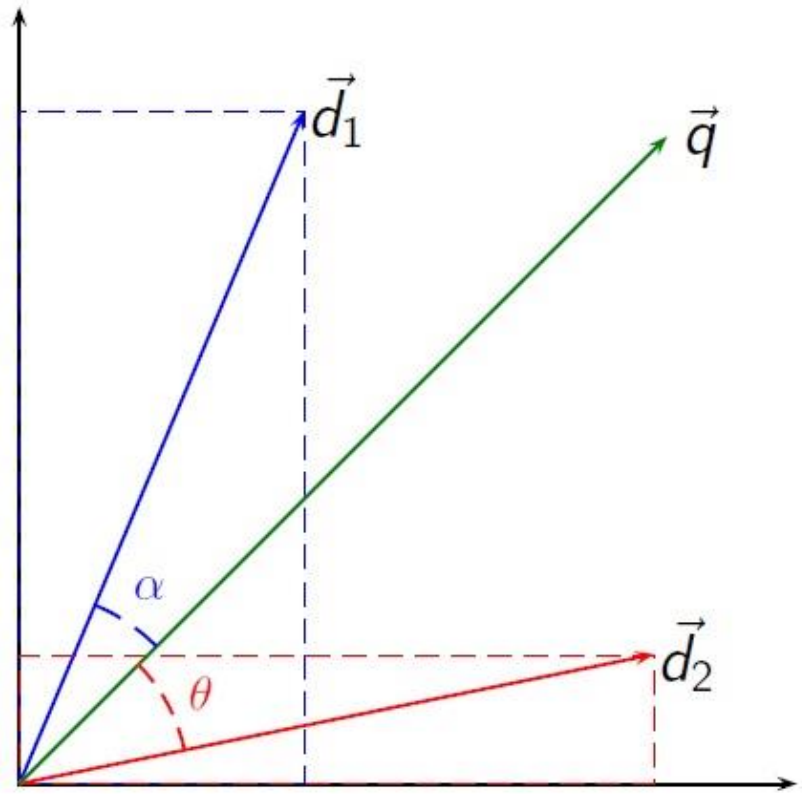
```
doc1 = [1, 1, 1, 0, 0]
```

- Now, use entire range [0, 1] to give more information

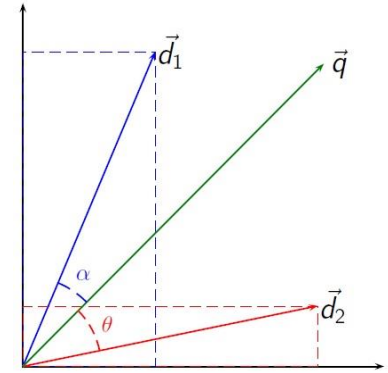
```
doc1 = [0.2, 0.1, 0.5, 0, 0]
```

TF-IDF and the vector space

- Vector lengths are now TF-IDF values
- Vectors that are "close" are still similar in meaning



Vector space similarity



- Vector space similarity is also called the cosine, or normalized inner product

- Cosine similarity measures the angle between two vectors. If the **angle** is small → documents are **similar**.

Vector space similarity

- Similarity of two docs is:

$$Sim(Di,Dj) = \square_{k=1}^{t} w_{ik} * w_{jk}$$

Normalized ahead of time, when computing term weights.

$$\mathrm{sim}(d_j, q) = \frac{\mathbf{d_j} \cdot \mathbf{q}}{\|\mathbf{d_j}\| \, \|\mathbf{q}\|} = \frac{\sum_{i=1}^{N} w_{i,j} w_{i,q}}{\sqrt{\sum_{i=1}^{N} w_{i,j}^2} \sqrt{\sum_{i=1}^{N} w_{i,q}^2}}$$

Not normalized ahead of time

Vector space similarity

- Euclidean dot product formula

$$\mathbf{A} \cdot \mathbf{B} = \|\mathbf{A}\| \, \|\mathbf{B}\| \cos\theta$$

$$\text{similarity} = \cos(\theta) = \frac{\mathbf{A} \cdot \mathbf{B}}{\|\mathbf{A}\| \|\mathbf{B}\|} = \frac{\sum_{i=1} A_i B_i}{\sqrt{\sum_{i=1}^{n} A_i^2} \sqrt{\sum_{i=1}^{n} B_i^2}}$$

Already "baked" into weight

$$Sim(Di, Dj) = \square_{k=1}^{t} w_{ik} * w_{jk}$$

Vector space summary

- User's query treated as short document
- Query is in same space as docs
- Easy to measure a doc's distance to query
- Extension of Boolean retrieval

TF-IDF strengths and weaknesses

- TF-IDF works great when
  - Topics are disjoint
  - Pages are high quality (ex. focused on specific topic)
- Fine for Wikipedia search
  - Likely used by Amazon product search
- General web search requires additional technique
  - Many relevant results
  - Many low-quality results

Summary

- Boolean retrieval includes or excludes results
- Vector space model: each document is a vector with one component for each term
- Vector space similarity
  - Documents that are close together are close in meaning
  - Query is a short document

Summary

- *Term frequency (TF)*: high value for common words in one document
- *Inverse Document Frequency (IDF)*: high value for rare words, low value for common words in collection of docs
- *tf-idf*: Combine TF and IDF to form the magnitude of components in vector